

The Kitchen of Life

I PUC · Ch 9 · Biomolecules

Starring Bindu 👧 & Prof. BAIO 🦉

A BAIO storybook by Narendra Sir · Biology Lecturer



PAGE 1 OF 8

What's Cooking?

Prof. BAIO ground a leaf with acid. "The **acid-soluble pool** holds small molecules like sugars and amino acids. The acid-insoluble part holds the big **macromolecules** — proteins, nucleic acids and polysaccharides." Lipids also land there, though they are small.

> 💡 **REMEMBER**
> **Biomolecules with molecular weight above ~1000 Da are macromolecules (lipids are an exception — they form membranes).**



Sweet Chains

Sugar molecules held hands using **glycosidic bonds** to make **polysaccharides**. Plants stored **starch**, animals stored **glycogen**, plant walls were built of **cellulose**, and insects wore armour of **chitin**. Dahlia stored **inulin**, a chain of fructose.

> 💡 **REMEMBER**
> **Starch gives a blue colour with iodine; cellulose does not.**



Twenty Builders

Twenty kinds of **amino acids** linked by **peptide bonds** built proteins. **Collagen** was the most abundant protein in animals, and **RuBisCO** was the most abundant protein in the whole biosphere.

💡 **REMEMBER**

A peptide bond forms between the –COOH of one amino acid and the $-NH_2$ of the next, releasing water.



PAGE 4 OF 8

Folding Up

A protein's **primary structure** is its sequence of amino acids. It coils into a helix — the **secondary structure** — then folds like a ball of wool — the **tertiary structure**. Some join with other chains in a **quaternary structure**, like haemoglobin.

💡 REMEMBER

Haemoglobin has four subunits: two α and two β.



Oils and Fats

Lipids were made of **glycerol** and **fatty acids**. Some fatty acids were saturated, some unsaturated. **Phospholipids** like lecithin were special — they built every cell membrane.

💡 **REMEMBER**

Oils are liquid at room temperature (more unsaturated); fats are solid.



The Code Letters

A **nucleotide** had three parts: a nitrogen base, a pentose sugar and a phosphate. **Purines** were adenine and guanine; **pyrimidines** were cytosine, thymine and uracil. Strung together, they made DNA and RNA.

💡 **REMEMBER**

Nucleoside = base + sugar · Nucleotide = base + sugar + phosphate.



The Magic Helpers

Enzymes were biocatalysts — they lowered the **activation energy** so reactions ran fast. Each had an **active site** where the substrate fitted like a key. Temperature, pH and substrate concentration changed their speed.

💡 **REMEMBER**

Enzymes are mostly proteins; ribozymes are RNA enzymes.



Enzyme Families

Prof. BAIO named six classes: oxidoreductases, transferases, hydrolases, lyases, isomerases and ligases. Some enzymes needed partners called **cofactors** — prosthetic groups, coenzymes or metal ions. A look-alike molecule could block the active site: a **competitive inhibitor**.

💡 **REMEMBER**

Malonate competitively inhibits succinic dehydrogenase.



KEY TERMS

Words you met in this story

Tap a word to jump to its page. Can you explain each one?

acid-soluble pool p.1 · macromolecules p.1 · glycosidic bonds p.2 · polysaccharides p.2 · starch p.2 · glycogen p.2 · cellulose p.2 · chitin p.2 · inulin p.2 · amino acids p.3 · peptide bonds p.3 · Collagen p.3 · RuBisCO p.3 · primary structure p.4 · secondary structure p.4 · tertiary structure p.4 · quaternary structure p.4 · Lipids p.5 · glycerol p.5 · fatty acids p.5 · Phospholipids p.5 · nucleotide p.6 · Purines p.6 · pyrimidines p.6 · Enzymes p.7 · activation energy p.7 · active site p.7 · cofactors p.8 · competitive inhibitor p.8



QUICK QUIZ

Do you remember?

1. The most abundant protein in the biosphere is:

- Collagen
- RuBisCO
- Haemoglobin
- Insulin

2. Purines are:

- A and G
- C and T
- C and U
- T and U

3. Animals store food as:

- Starch
- Cellulose
- Glycogen
- Inulin

4. Enzymes speed up reactions by:

- Raising temperature
- Lowering activation energy
- Adding energy
- Changing pH

5. Malonate inhibits succinic dehydrogenase as a:

- Cofactor
- Coenzyme
- Competitive inhibitor
- Substrate

Answers: 1-RuBisCO · 2-A and G · 3-Glycogen · 4-Lowering activation energy · 5-Competitive inhibitor